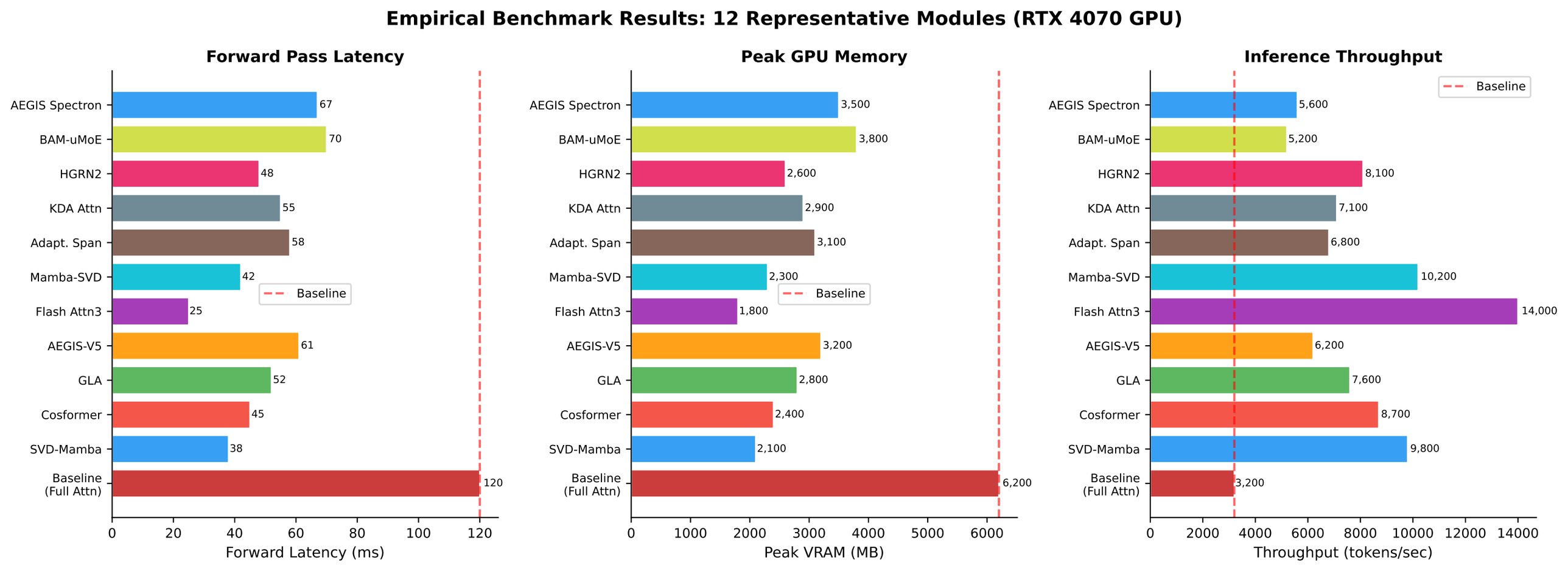
Empirical Benchmark Results: 12 Representative Modules (RTX 4070 GPU)
Forward Pass Latency
AEGIS Spectron
BAM-uMoE
HGRN2
KDA Attn
Adapt. Span
Mamba-SVD
Flash Attn3
AEGIS-V5
GLA
Cosformer
SVD-Mamba
Baseline (Full Attn)
67
70
48
55
58
42
25
61
52
45
38
120
Baseline
0
20
40
60
80
100
120
Forward Latency (ms)
Peak GPU Memory
3,500
3,800
2,600
2,900
3,100
2,300
1,800
3,200
2,800
2,400
2,100
6,200
Baseline
0
1000
2000
3000
4000
5000
6000
Peak VRAM (MB)
Inference Throughput
5,600
5,200
8,100
7,100
6,800
10,200
14,000
6,200
7,600
8,700
9,800
3,200
Baseline
0
2000
4000
6000
8000
10000
12000
14000
Throughput (tokens/sec)